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Q1. (if Statement)

```
age = int(input("Enter your age: "))
```

```
if age >= 18:
```

```
    print("You are eligible to vote.")
```

Q2.(if-else Statement)

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:
```

```
    print("Even Number")
```

```
else:
```

```
    print("Odd Number")
```

Q3.(if-elif-else Statement)

```
marks = int(input("Enter your marks: "))
```

```
if marks >= 90:
```

```
    print("Grade A - Excellent")
```

```
elif marks >= 75:
```

```
    print("Grade B - Good")
```

```
elif marks >= 60:
```

```
    print("Grade C - Average")
```

```
elif marks >= 40:
```

```
    print("Grade D - Pass")
```

```
else:
```

```
    print("Fail")
```

Q4.(For Loop with range)

```
print("Numbers from 1 to 30:")
```

```
for i in range(1, 31):
```

```
    print(i)
```

```
print("\nOdd numbers from 1 to 50:")
```

```
for i in range(1, 51, 2):
```

```
    print(i)
```

```
total = 0
```

```
for i in range(1, 16):
```

```
    total += i
```

```
print("\nSum of numbers from 1 to 15 =", total)
```

Q5.(While Loop)

```
num = 1
```

```
while num <= 20:
```

```
    print("Number =", num, " Cube =", num ** 3)
```

```
    num += 1
```

Q6.(While Loop - User Controlled)

```
total = 0
```

```
while True:
```

```
    num = int(input("Enter a positive number (0 or negative to stop): "))
```

```
    if num <= 0:
```

```
        break
```

```
total += num  
print("Total Sum =", total)
```

Q7.(Nested if Statement)

```
temperature = int(input("Enter temperature: "))  
is_raining = input("Is it raining? (yes/no): ")  
if temperature > 30:  
    if is_raining == "yes":  
        print("Hot and rainy, carry umbrella.")  
    else:  
        print("Hot day, wear light clothes.")  
elif temperature < 15:  
    if is_raining == "yes":  
        print("Cold and rainy, wear jacket and take umbrella.")  
    else:  
        print("Cold day, wear warm clothes.")  
else:  
    print("Weather is pleasant. Have a nice day!")
```

Q8.(For Loop + if-elif-else)

```
for i in range(1, 41):  
    if i % 3 == 0 and i % 5 == 0:  
        print("FizzBuzz")  
    elif i % 3 == 0:
```

```
    print("Fizz")
elif i % 5 == 0:
    print("Buzz")
else:
    print(i)
```

Q9.(Menu Driven Program)

while True:

```
    print("\n===== Simple Calculator =====")
    print("1. Add two numbers")
    print("2. Subtract two numbers")
    print("3. Multiply two numbers")
    print("4. Divide two numbers")
    print("5. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 5:
        print("Thank you! Program Ended.")
        break
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    if choice == 1:
        print("Addition =", num1 + num2)
    elif choice == 2:
        print("Subtraction =", num1 - num2)
```

```
elif choice == 3:

    print("Multiplication =", num1 * num2)
elif choice == 4:

    if num2 != 0:

        print("Division =", num1 / num2)

    else:

        print("Error: Division by zero is not allowed.")
else:

    print("Invalid choice! Please enter a number from 1 to 5.")
```

Q10.(Debugging)

```
num = int(input("Enter a number: "))

if num > 100:

    print("Large number")

elif num > 50:

    print("Medium number")

else:

    print("Small number")

count = 1

while count < 10:

    print(count)

    count += 1
```